Chapter5: Virtualization Features

1. In FusionCompute if a VM is in the running or stopped state, users can increase or decrease the number of vCPUs of the VM as required. **(False)**

2. Huawei FusionCompute allows users to add hosts in batches using template. **True**

3. Which of the following about virtual network computing VNC are true
   A. VNC must support the Linux OS.
   B. VNC is a lightweight remote control computer software.
   C. A controlled computer can function as a VNC server only when the corresponding software is installed on it.
   D. The host that initiates a control request can function as a VNC server only when the corresponding software is installed on it.

4. Which of the following is **not** included in the snapshot management functions?
   A. Creating snapshot
   B. Modifying a snapshot
   C. Deleting a snapshot
   D. Copying a snapshot

5. Which of the following is false about snapshots?
   A. A consistency snapshot will be created if disk I/O pressure is high. You are advised to create consistency snapshot when pressure is low.
   B. After a snapshot is generated,incremental disks take space of the data store where disks are located.to prevent storage resource insufficiency control the number of snapshots.
   C. Another snapshot can be created for a VM when a snapshot is being created for the VM.
   D. Snapshots don't store information about pass through devices such as USB/GPU/SSD.

6. Which of the following is false about snapshots?
   A. Snapshots can be used to store VM settings and VM disk data.
   B. Snapshots can be used to store VM memory status.
   C. Multiple snapshots can be created for a VM. If a VM is restored using a snapshot, the later snapshots become invalid.
   D. Only one memory snapshot can be created for all VMs on a host at a time.

7. During the creation of the snapshot, users can perform other operations on the VM. **(False)**
8. Which of the following scenarios are snapshots inapplicable to?
   A. Critical service change
   B. Service batch installation
   C. Destructive testing of services
   D. Long-term storage of service data

9. Which of the following statements about snapshot-based VM restoration is false?
   A. A vm can have multiple snapshot
   B. If the port group where VM NICs belong is deleted after a snapshot is created for the VM, the deleted port group will be restored when the VM is restored using the snapshot.
   C. If the disk mode of the VM is independent and persistent or independent and non-persistent , the data on the disk will not be restored when the VM restored using snapshot.
   D. A created snapshot of a VM can be used to restore the VM data to the state when the snapshot was created.

10. Which of the following statement is **false** about memory over commitment?
   A. The VM performance deteriorates on a host that has the memory over commitment function enabled.
   B. The memory over commitment function can be disabled only when a memory over commitment ratio of all hosts in the cluster is less than or equal 100%.
   C. After memory over commitment is disabled, the total memory size of all VMs is less than or equal to the total available physical memory size
   D. Administrators can enable or disable the memory over commitment at any time.

11. Which statements about live migration conditions is false?
   A. The source host and destination host communicate with each other properly
   B. The destination host has sufficient resources
   C. The VM is running
   D. The type of destination storage device is different from of the source storage device.

12. Which is false about live migration application scenario?
   A. If a host is faulty or overloaded ,the VMs running on it can be migrated to another host to prevent service interruption.
   B. If the service load is too light on multiple hosts, the VM can be migrated to a smaller number of hosts which decrease the power consumption.
   C. VMs at a graphics-capable workstation can be live migrated after being bound to an image processor.
   D. During host maintenance. VMs running on the host can be migrated to other hosts to ensure normal service running.

13. Which statements about live migration and cold migration is false?
   A. Cold migration is to migrate a stopped VM from one host to another. During the migration, services are interrupted.
   B. Live migration is to migrate a running VM from one host to another. During the migration, services are not interrupted.
   C. The reliability test in performance tests includes the tests on the VM migration.

D. Memory data and disk data are migrated during live migration and cold migration.

14. Which of the following QoS parameters can set for VMs on FusionCompute?
    A. Memory
    B. CPU
    C. GPU
    D. Disk IOPS
    E. Network

15. (Multiple Answer) Which of the following aspects reflect that the virtualization technology of cloud computing improves the resource usage?
    A. VM resource adjustment
    B. Memory overcommitment
    C. Improving server usage
    D. Automatic application deployment

16. (Multiple Answer) Huawei provides end-to-end cloud computing solutions covering hardware and software. Which of the following hardware components is optional in the Huawei cloud computing solutions?
    A. Server
    B. Load balancer
    C. Firewall
    D. TCs

17. (Multiple Answer) Which of the following functions are based on Huawei Virtual Image Management System (VIMS)?
    A. Thin provisioning
    B. VM snapshot
    C. Storage migration
    D. Linked cloning

18. VM Live migration is to migrate VMs from one host to another in the same cluster when the VM is powered off**. (False)**

19. (Single Answer) In the Huawei FusionSphere solution, if HA needs to be triggered to migrate VMs to other servers in the event of a server failure, which of the following conditions **is not mandatory**?
    A. The hosts must be in the same cluster.
    B. HA has been enabled for the cluster accommodating the host.
    C. The hosts are connected with the same shared data store.
    D. VMs that require the HA function use the same network IP address segment.

20. (single) Which of the following will not cause a VM migration failure on FusionCompute?
    A. DPM is not enabled.
    B. The VM is bound to the host.
    C. The destination host is unavailable.
    D. The source host is in maintenance mode.

21. Assume that three VMs A, B and C run on the host that uses a single core CPU whose clock speed is 2.8 GHz,their respective quotas are set to 100, 200 and 400 and their respective CPU reservation values set to 700 MHz, 0 MHz and 0MHz. What is the computing capability of VM C when the three VMs running at full CPU load?
    A. 700 MHz
    B. 300 MHz
    C. 1400 MHz
    D. 600 MHz

22. (Single-Answer Question) Which of the following batch operations is not supported by FusionCompute?
    A. Adding hosts in batches
    B. Binding network ports in batches
    C. Associating storage resources in batches
    D. Migrating VMs in batches

23. (Multi-Answer Question) Which of the following features does FusionCompute support?
    A. VM HA
    B. Dynamic Resource Scheduling (DRS)
    C. CPU QoS
    D. VM live migration from the hosts using AMD CPUs to hosts using INTEL CPUs.

24. Which of the following statements about memory overcommitment is false?

    A. Memory overcommitment technologies include memory ballooning, memory swapping, and memory sharing
    B. Memory overcommitment technologies allow the total memory size of VMs to be greater than the total memory size of the physical server.
    C. After memory overcommitment is enabled, the memory overcommitment ratio is inversely proportional to the actual memory usage of each VM.
    D. Memory overcommitment technologies allow the used memory size of VMs to be greater than the total memory size of the physical server.

25. （Single）In which scenario should the memory overcommitment ratio not be increased?
    A. The VM memory usage is low.
    B. The memory overcommitment ratio reaches 150% in commercial scenarios.
    C. The customer requires that the maximum allowed number of VMs be deployed on a single host while ensuring performance.
    D. Customers' hardware procurement costs need to be reduced.

26. In Huawei FusionCompute, which of the following statements about CPU resource QoS is false?
    A. CPU quota defines the proportion based on which CPU resources to be allocated to each VM when multiple VMs compete for physical CPU resources.

B. CPU limitation defines the priority based on which CPU resources to be allocated to each VM when multiple VMs compete for physical CPU resources.

C. CPU reservation takes effect only when VMs compete for resources. If the CPU resources are sufficient, a VM can exclusively use physical CPU resources on the host if required.

D. CPU reservation defines the minimum CPU resources to be allocated to each VM when multiple VMs compete for physical CPU resources.

27. （Multiple）In Huawei FusionCompute, if a VM is bound to a host, which of the following functions will be affected?
A. Snapshot
B. DRS
C. Hot migration
D. Memory hot-add

28. （T or F）Huawei FusionCompute is able to check the usage of global physical resources during VM running, calculate the optimal host for the VM based on the intelligent scheduling algorithm, and enable the VM to run on the optimal host by means of live migration, thereby improving the overall service experience. **True**

29. （Single）In Huawei FusionCompute, which of the following statements about the IMC function is false?
A. You are advised to enable the IMC function when the CPU models of CNA hosts in the cluster are inconsistent.
B. If the IMC function is enabled, VMs cannot be migrated between hosts with different CPU models. Determine whether to enable the IMC function based on the site requirements.
C. IMC only supports the VM migration from an earlier CPU generation to a later CPU generation.
D. You are advised to enable the IMC function when the memory sizes of CNA hosts in the cluster are inconsistent.

30. O&M engineers can migrate VMs on a physical server to be upgraded to other servers to prevent service interruption during system maintenance performed by carrier. **True**

31. Which of the following about NUMA is false?

A. Guest NUMA presents the NUMA topology to VMs to enable the guest OS to allocate resources to applications running on it based on logical NUMA nodes,thereby improving application performance.
B. Host NUMA provides the CPU load balancing mechanism to eliminate the VM performance bottleneck due to unbalanced CPU resource allocation.
C. Host NUMA ensures that VMs access only the local physical memory which reduces memory access latency which improve VM performance.

D. With NUMA, the speed of CPU access to memory on the same node is the same as that of access to memory on another node.

32. Which of the following are included in **virtual** NUMA?

A. Topology presentation: NUMA is identified in the VM to make NUMA optimization of the guest OS and applications effective.

B. Initial placement: based on the VM NUMA topology, select a physical node to place the vCPU and memory, so that the relationship between the vCPU and memory on the vNode is consistent with the relationship between CPU and memory on the physical node.

C. Load balancing: During the scheduling, the association between nodes and the mapping between vNodes and physical nodes are considered to maximize the speed of vCPU access to local memory.

D. Dynamic migration: when the affinity relationship between the vCPU and physical node changes, memory on the corresponding vNode will be migrated.

33. Which of the following about the purpose of enabling the NUMA function is true?

A. Optimize the memory access performance of VMs.
B. Maximize the CPU computing capability of the physical host.
C. Maximize the vCPU computing capability of VMs
D. Maximize the memory over commitment ratio of the physical host.

34. NUMA is a multiprocessor –based computer memory design. The memory access time depends on the relative location of memory to the processor. In NUMA, the speed of a processor to access its own local memory is faster than that to access a non-local memory (memory located on another processor or memory shared by processor). **True**

35. Which of the following statements about NUMA Topology Adjustment switch is false?

A. If the optimal NUMA topology cannot be calculated, the VM behavior is consistent with the switch status.

B. After the switch is set to OFF, interleaving access occurs on NUMA nodes during memory access, resulting in performance loss.

C. After the switch is set to ON, the system sets VM NUMA affinity and physical NUMA affinity during automatic adjustment to maximize VM memory access performance.

D. After this switch is set to ON, VMs may not be scheduled to the specified nodes, and CPUs and memory are evenly allocated to bound nodes during manual adjustment.

36. In Huawei FusionCompute, snapshots can be taken for computing Node Agent (CNA) hosts. **False**

37. In Huawei FusionCompute, which of the following statements about CPU resource QoS is false?
    A. Adjust the CPU reservation, quota and limitation to control the CPU usage of VMs
    B. CPU reservation is used to ensure the minimum VM computing capability ,that is, the minimum number of resources that can be used.
    C. CPU quota is used to balance and adjust resources based o9n the relative CPU quotas of multiple VMs when computing resources are insufficient.
    D. CPU limitation is used to specify the maximum number of VMs that can be run on a host.

38. In Huawei FusionCompute, which of the following functions will be affected after a USB device is attached to a VM?

    A. Restarting a VM
    B. VM live migration
    C. Cluster scheduling
    D. VM resource capacity expansion

39. In Huawei FusionCompute, the admin binds the two NICs of the CNA host in **Active/standby** mode. After the binding ,the data transmission rate is equal to the sum of the transmission rates of the two NICs. **False**

40. In Huawei FusionCompute, after a user logs in to a VM using VNC, operations on the VM fail, for example, files cannot be opened. Which is the most possible cause of this problem?

    A. The management link is unstable
    B. The service link is unstable
    C. The storage link is unstable
    D. The VM is damaged

41. In Huawei FusionCompute, when a compute node is powered off, the system can start HA-enabled VMs on the compute node from other compute nodes. Which of the following **is not a prerequisite** for using this function?

    A. HA-enabled VMs on a compute node use IP addresses on the same network segment.
    B. The VM is not bound to the host
    C. HA is enabled for the cluster to which the compute node belongs.
    D. The source and destination compute nodes share the same data store

42. In Huawei FusionCompute, VM flavors cannot be modified during VM cloning. **False**

43. In Huawei FusionCompute, VMs can be deployed in batches using a template. **True**

44. Which of the following about cluster, host and VM In Huawei FusionCompute, is **false**?
    A. Hosts can move between different clusters.
    B. VMs can be migrated across clusters
    C. Whether multiple hosts belong to the same cluster depends on whether they use the same shared storage
    D. A cluster consists of multiple hosts and their associated storage and network resources.

45. In Huawei FusionCompute, which of the following are the methods for creating VM?
    A. Importing a VM from other virtualization platforms
    B. Deploying a VM using a template
    C. Cloning a VM from an existing VM
    D. Creating a new VM

46. In Huawei FusionCompute, if a host in a cluster becomes faulty, VMs on the host are automatically migrated to other hosts in the cluster. **False**

47. In Huawei FusionCompute, which of the following is not a method to create template?

    A. Converting a VM to a VM template
    B. Converting a snapshot to a VM template
    C. Cloning a VM to a VM template
    D. Cloning a VM template to a VM template

48. The resources in a cluster consists of hosts and associated storage and network resources. **True**

49. In Huawei FusionCompute, which of the following statements about VM storage live migration is **false**?
    A. Storage live migration can be performed between different datastrores.
    B. Storage live migration can be performed between different storage resources
    C. Storage live migration can be performed between different storage devices
    D. Storage live migration can be performed between different virtual disks.

50. In Huawei FusionCompute, memory and CPU resources used by VMs can only be provided by the same cluster. Therefore, the maximum number of available computing resources of VM cannot exceed the total number of resources that can be provided by all host in the cluster. **False**

51. What operations can administrator perform on the cluster?
    A. Changing cluster name
    B. Exporting the cluster to a local directory
    C. Adding cluster description
    D. Moving a cluster to a specified folder

52. In Huawei FusionCompute, Users can create VMs by importing a VM template file. Which of the following file formats is not supported?

A. .ova
B. .qcow2
C. .vmdk
D. .ovf

53. In Huawei FusionCompute, which **is not a** restriction for VM live migration?
A. The shared disk cannot be attached
B. The VM cannot used raw device mapping RDM datastores
C. The VM must be in the Running state.
D. The destination host is not in maintenance mode.

54. In Huawei FusionCompute, which of the following statements about the factors affecting computing resource scheduling is false?

A. When both CPU and memory are configured as the factors, the scheduling policy is triggered only when the CPU and memory conditions are met.

B. When CPU is configured as the only factor, the scheduling policy is triggered when the CPU conditions are met.

C. When memory is configured as the only factor, the scheduling policy is triggered when the memory conditions are met.

D. When both CPU and memory are configured as the factors, the scheduling policy is triggered when either the CPU or the memory conditions are met.

55. Which of the following about Huawei FusionCompute HA is false?
A. This function enables a VM to reboot whenever the VM encounters with a failure.
B. If VM data is stored in the shared storage, the unsaved data will not be lost if a fault occurs.
C. With HA enabled the system periodically detects the VM status. If detecting a VM fault caused by physical server fault, the system automatically migrates the VM to another physical server and restarts the VM to allow rapid restoration of the VM services.
D. Administrators can set different HA policies based on importance of VMs.

56. Which is **false** about templates?
A. After a template is converted to a VM, the template will no longer exist.
B. Parameters can be customized when a VM is deployed using a template
C. A template can be used across virtualization platforms.
D. After a VM cloned as a template, the VM still exists.

57. Which is **false** about template format?
A. The VM template formats include: .ova and .ovf
B. After a template in .ova format is exported, the template contains only one file.
C. The .ovf file is the VM description file and the .vhd file the VM disk file
D. An .ovf template only contains one .ovf file and one .vhd file

58. Which is **false** about creating templates on Huawei FusionCompute?

A. Convert a VM to a VM template: All parameters of the template are from the VM. The VM does not exist after the conversion.

B. Clone a VM to a VM template: All parameters of the template are from the VM, but some parameters can be modified during the cloning process. The VM is still available after the cloning complete

C. Clone a VM template to a VM template: All parameters of the new template are from the origin VM template, but some parameters can be modified during the cloning process. The original VM template is still available after the cloning is complete.

D. The VM used for creating a template can be in the Running state.

59. When users use a template to quickly deploy a VM, the generated VM configuration information is the same as that in the template. **False**

60. When users use a template to quickly deploy a VM, they don't need to install the OS or performed related configurations. **True**

61. Templates can be used to quickly deploy a VM. Currently, users can create templates only by converting or cloning VMs. **False**

62. In the Huawei FusionCompute, users can configure DRS rules to enable multiple VMs to run only in the specified host group. **True**

63. In the Huawei FusionCompute, which of the following deployment modes is recommended to improve VRM reliability?
    A. Single node deployment
    B. Active/Standby deployment
    C. Active/Active deployment
    D. Hybrid deployment

64. Which method is used for installing VRM VMs using FusionCompute installer?
    A. Automatic installation of an iso file
    B. Importing a template file to a VM
    C. Automatic installation using PXE
    D. Automatic VM creation and iso file mounting

65. In practice, resources must be reserved for hosts in a cluster to ensure that VM HA can be performed properly. **True**

66. VM live migration is a perquisite for implementing VM HA. **False**

67. Core services of an enterprise are deployed on multiple VMs in cluster mode, High availability (HA) policies can be configured for the VMs to improve services reliability. **True**

68. In FusionCompute, which can be performed by the system if a Blue Screen of Death (BSOD) occurs on a Windows VM?
    - **A. No processing is performed**
    - **B. Restarting the VM**
    - C. Restoring the VM using a snapshot
    - **D. Performing VM HA**

69. In FusionCompute, after the automatic computing resource scheduling function is enabled, which are possible causes for VM migration?
    - **A. The system automatically migrates a VM according to the scheduling policy.**
    - **B. A BSOD occurs on a Windows VM and the handling policy is HA.**
    - **C. The administrator manually migrates a VM.**
    - D. Applications on a VM break down

70. In FusionCompute, which are trigger conditions for VM HA?
    - **A. The host where the VM is located is faulty.**
    - B. Applications on the VM break down.
    - **C. A BSOD occurs on Windows VMs.**
    - **D. The datastore where the VM is located is faulty.**

71. In FusionCompute, which are **true** about VM HA?

    - A. After the HA function is enabled in a cluster, if a host in the cluster becomes faulty, VMs on the host can be rebuilt on other hosts in the cluster.
    - **B. When the system detects BSOD on a VM and the handling policy configured for this error is HA, the system creates the VM on a normal compute node.**
    - **C. VM HA can quickly recover services if a fault occurs.**
    - D. If a fault occurs, HA-enabled VMs can be successfully rebuilt on other compute nodes as long as there are reserved resources in the cluster.

72. In FusionCompute, the Distributed Resource Scheduling (DRS) migrates VMs based on the following factors:
    - **A. CPU**
    - **B. Memory**
    - C. I/O
    - D. Network traffic

73. In FusionCompute, Distributed Power Management (DPM) depends on DRS. **True**

74. Huawei FusionCompute **DRS** rules don't take effect for VMs with **USB** devices attached. **True**

75. Which of the following requirements can be met by using Huawei FusionCompute DRS rules?
    - **A. A VM runs on a specified host.**
    - B. If a host in a cluster is unavailable, VMs running on the host are automatically migrated to other hosts.
    - C. If a BSOD occurs on a Windows VM, the system automatically restart the VM.
    - D. If the VM configuration doesn't meet service requirements, the management system automatically allocates more CPU resources to the VM.

76. Which of the following operation can be performed even if the tool is not installed on the VM?
    - A. Restarting VM
    - **B. Forcibly stopping the VM**
    - C. Stopping VM
    - D. Converting VM to template

77. Any of intelligent memory over commitment function can be enabled and disabled based on customer requirements. **False**

78. Memory over commitment is a resources allocation solution. It reduces costs and improves VM density on a single host at the cost of performance. **True**

79. In FusionCompute, where can the memory over commitment function be enabled?
    - A. OS
    - B. VM
    - C. Host
    - **D. Cluster**

80. Which is **false** about memory over commitment?
    - A. After memory overcommitment is enabled, the total memory size of all VMs created by users can exceed the physical memory.
    - B. Enabling memory over commitment improves VM deployment density on physical hosts
    - **C. When memory over commitment is enabled, a higher over commitment ratio indicates higher VM performance.**
    - D. If the memory usage of a user VM is high, the memory over commitment ratio can be adjusted to optimize the memory usage.

81. In which scenario memory over commitment function recommended to be **disabled**?
    - A. Common office VMs.
    - B. Telecom customer service center
    - C. Bank customers

D. Graphics-capable workstations

82. Memory swapping means the external storage is virtualized into memory for VM to use. Data that is not used temporarily is stored in external storage. If the system needs such data, the system exchanges the data with the data reserved in memory. **True**

83. Huawei FusionCompute is able to check the usage of global physical resources during VM running, calculate the optimal host for the VM based on intelligent scheduling algorithm, and enable the VM to run on optimal host by mean of live migration, thereby improving the overall service experience. **True**

84. In Huawei FusionCompute, if VMs need to be migrated between CNA hosts with different CPU models, which functions must be enabled?
    A. Cluster I/O ring adaption
    B. Cluster HANA optimization
    C. Cluster GuestNUMA policy
    D. Cluster IMC policy

85. In Huawei FusionCompute, which is not a QoS parameter of CPU resources?
    A. CPU quota
    B. CPU limitation
    C. CPU reservation
    D. CPU over commitment

86. If a VM of a customer requires a CPU clock speed of at least 2 GHz, which of the following methods should be used?
    A. CPU quota
    B. CPU limitation
    C. CPU reservation
    D. CPU over commitment

87. CPU resource QoS and memory overcommitment are used to allocate resources appropriately based on different policies. **True**
88. Which is false about CPU QoS and memory QoS?
    A. CPU QoS of a VM is used to ensure that the computing resources of the VM are properly allocated.
    B. CPU quota and CPU reservation take effect only when VMs compete for computing resources
    C. Memory QoS is used to ensure that key service VMs can obtain sufficient memory resources
    D. Memory QoS can be used only when memory overcommitment is enabled

89. Memory overcommitment is the perquisite for implementation of memory resource QoS.
False

90. An e-commerce service VM requires memory of at least 8 GB to cope with service traffic bursts during holidays. Which can be used to meet this requirement?

A. Memory overcommitment
B. Memory QoS
C. VM live migration
D. Rapid VM deployment

91. When users use a Windows template to quickly deployed VMs, the generated SIDs may be the same. As a result, some special software may fail to be installed. **True**

92. In Huawei FusionCompute, which is **true** about role?
A. A role can have one or more operation permissions.
B. A user can attached to only one role
C. If a user attached to multiple roles, the operation permissions of the user depend on the role with the highest priority.
D. the operation permissions of a role cannot be modified